$$\rho = \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{2} \end{pmatrix}. \tag{7.24}$$

The trace of our matrix is 1. And our density matrix is done.[8]

Exercise 7.7: Use Eq. 7.24 to calculate ρ^2. How does this result confirm that ρ represents an entangled state? We'll soon discover that there are other ways to check for entanglement.

Exercise 7.8: Consider the following states:

$$\begin{aligned} |\psi_1\rangle &= \frac{1}{2}\Big(|uu\rangle + |ud\rangle + |du\rangle + |dd\rangle\Big) \\ |\psi_2\rangle &= \frac{1}{\sqrt{2}}\Big(|uu\rangle + |dd\rangle\Big) \\ |\psi_3\rangle &= \frac{1}{5}\Big(3|uu\rangle + 4|ud\rangle\Big). \end{aligned}$$

For each one, calculate Alice's density matrix and Bob's density matrix. Check their properties.

7.7 TESTS FOR ENTANGLEMENT

Suppose I gave you a wave function

$$\psi(a, b)$$